The Athlete's Wrist: Ulnar-Sided Pain

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Abstract

Ulnar-sided wrist pain is one of the most common symptoms in athletes of baseball, racket sports, golf, and wrestling where there is frequent use of the hands as well as in soccer and running, where hand use is minimal. Compared with all wrist injuries, ulnar-sided wrist injury is a relatively serious condition for athletes because it plays an important role in performing a strong grip and in the rotation of the forearm.

Ulnar-sided wrist pain in athletes can be related to acute trauma or chronic overuse. Acute trauma can lead to bone fractures and sprains/tears of ligaments. Repetitive mechanical stresses to tendons, ligaments, and the joint structures can lead to tendinitis or osteoarthrosis.

Diagnosis of the ulnar-sided wrist pain is challenging both for hand surgeons and radiologists because of the small and complex anatomy. In the present article, we discuss mechanisms of wrist injury, sports-specific ulnar-sided wrist injuries, and the differential diagnosis of ulnar-sided wrist pain.

Keywords
- athlete
- sports
- ulnar-sided wrist pain
- MRI
- radiograph

Many athletes experience wrist injuries. Wrist injuries can occur during sports such as baseball, tennis, golf, wrestling, gymnastics, and so on, where hands are often used. Wrist injuries also occur in sports such as soccer and running where the hands are infrequently used. Wrist pain can have many etiologies, but ulnar-sided wrist injury and pain is relatively more serious because the ulnar-sided structures are vital in performing a strong grip and in forearm rotation.

Proper diagnosis of an athlete's injury is essential because an inappropriate treatment may result in the delay of return to activity and in reduction of performance. Furthermore, structures of the wrist are so small and complex that general orthopedic surgeons, who are not specialized in hand surgery, sometimes misdiagnose disorders such as intercarpal ligament injury or overlook an occult fracture. It is important to obtain precise imaging assessment using radiographs, computed tomography (CT), and/or MRI to assist orthopedic hand surgeons in arriving at an accurate diagnosis of ulnar-sided wrist pain. This article reviews the mechanisms of ulnar-sided wrist injury and their characteristic radiological findings, with a special focus on ulnar-sided wrist pain.

Mechanism of Injury to the Wrist

Mechanisms of sports-related ulnar-sided wrist injury can be roughly divided into four motions: throwing, weightbearing, twisting, and impaction. Overuse injuries due to throwing may occur as a result of repetitive flexion/extension and radial/ulnar deviation. Wrist injuries due to throwing are prevalent in overhead activities such as baseball, softball, racket sports, water polo, and handball. Tendinitis of the extensor carpi ulnaris (ECU) and the flexor carpi ulnaris (FCU) is frequently seen in throwing-related injuries. Weightbearing requires the athlete to endure repetitive, high-compressive forces at the wrist joint. Wrist injuries due to weightbearing are commonly seen in gymnastics, wrestling, and weightlifting. Under these circumstances, the triangular fibrocartilage complex (TFCC) is subjected to damage. Twisting, that is, forceful rotation of the wrist, can occur in all athletic activities and affect joint stability. Most twisting injuries result in the impairment of the distal radioulnar joint (DRUJ) or ulnar carpus. Wrist injuries due to impact are characterized by direct impact or fall onto the wrist. These

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injuries are mainly seen in the contact sports such as wrestling, football, rugby, basketball, and soccer as well as noncontact sports such as baseball and running. Ulnar-sided wrist injuries due to impact force include hamate fracture, ligamentous injury, scapholunate dissociation, lunotriquetral dissociation, and TFCC tear.

Sport-Specific Ulnar-Sided Wrist Injuries

Baseball

Most wrist injuries in baseball are caused by a direct hit with the ball.[1] Injuries due to collision with another player or objects such as an outfield fence occur less frequently. Sprains account for 33% of the injuries, and fractures represent 46%.[1] One of the most common carpal fractures resulting in ulnar-sided wrist pain is a fracture of the hook of the hamate. Baseball batters fracture the hook of the hamate of the nondominant hand by the counterforce to the butt end of the bat while hitting the ball foul rather than a direct hit with the ball. Baseball players sliding into base with hyperdorsiflexion of the wrist may injure the TFCC. Overuse injuries, such as tendinitis of the ECU and FCU, may occur as a result of throwing. Other overuse injuries, including ECU subluxation or dislocation, may result from batting.

Racket Sports

The incidence of upper extremity injuries in tennis is 20 to 45%.[2] Of these, ~7.7% involve the wrist.[3] The wrist is in extension position at the time of impact on all strokes and in ulnar deviation at impact in the serve. Advanced players demonstrate more extension at impact than developing amateur players. Ulnar deviation on impact may account for the frequency of ulnar-sided wrist injuries. Ulnar-sided wrist pain in racket sports players includes TFCC injury, ECU tendinitis, ECU subluxation, ulnar impaction syndromes, and fracture of the hook of hamate. In particular, ulnar impaction syndrome, also known as ulnar abutment, is a characteristic injury of tennis players. Symptoms of ulnar impaction syndrome are similar to those of TFCC tear because ulnar abutment may result in TFCC injury as a consequence of continuous TFCC compression.

Golf

McCarroll et al reported that the left wrist was the most common site of injury in professional golfers, specifically 24% of all injuries.[4] They also reported that the hand and wrist were injured in 20% of amateur golfers.[4] Most right-handed golfers' injuries to the left wrist include De Quervain tenosynovitis and radial impaction syndrome, which are radial-sided injuries. Common ulnar-sided injury in golfers is fracture of the hook of the hamate, which occurs during impact. TFCC injuries may also occur when the player hits the ground with the golf club.

Gymnastics

Gymnastics also causes ulnar-sided wrist pain in athletes. In gymnastics, the upper extremity has the unique feature of being converted into a weightbearing system. Because of the nature of most gymnastics events, the wrist is subject to stresses that predispose it to higher injury rates than the elbow or shoulder. The load commonly applied to the wrist is compression in a dorsiflexed position. Radial and ulnar deviation while in this position results in an extensive array of wrist injuries.[5] Between 46% and 87% of elite gymnasts complain of wrist pain during their career.[6–8] Ulnar impaction syndrome is one of the most common ulnar-sided wrist pains in gymnasts.

Differential Diagnosis for Ulnar-Sided Wrist Pain

Ulnar Styloid Fracture

Although an ulnar styloid fracture can be isolated, it is often accompanied by a distal radius fracture. Frykman reported that ulnar styloid fractures occurred in ~61% of distal radius fractures.[9] An isolated styloid fracture often results from a direct blow, whereas combined styloid and distal radius fractures generally result from a fall accident. Ulnar styloid fractures are easily diagnosed by posteroanterior (PA) radiographs (►**Fig. 1**) and classified into four groups related to the site of fracture: tip, middle, base, and base oblique (►**Fig. 2**).[10] The base and base oblique fractures involve the triangular ligament, also known as the distal radioulnar ligament. This can lead to DRUJ stability; therefore, it is important to obtain an MRI to evaluate for ligamentous injury when base or base oblique fractures are identified on radiographs. In contrast, fractures through the tip or middle of the ulnar styloid do not lead to DRUJ instability and usually result in a good prognosis.



Figure 1 Posteroanterior view of radiograph of the wrist in 28-year-old male football player. There is an obvious fracture at the base of ulnar styloid with distal radius fracture (arrow).

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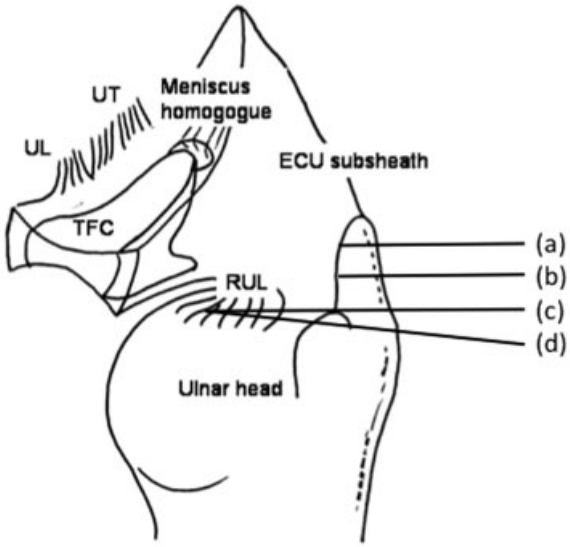


Figure 2 Classification of ulnar styloid fracture: (a) tip, (b) middle, (c) base, and (d) base oblique of the styloid. Base and base oblique fractures involve the triangular ligament, which is concerned with distal radioulnar joint stability. ECU, extensor carpi ulnaris; RUL, radioulnar ligament; TFC, triangular fibrocartilage complex; UL, ulnolunate ligament; UT, ulnotriquetral ligament.



Figure 3 Radiograph of the left hamate hook fracture in a 26-year-old male baseball player. Fracture line (arrows) is apparent in (a) carpal tunnel view and (b) supinated oblique view with the wrist-extended position.

Hamate Fracture

There are two main types of hamate fractures; one is fracture of the body and the other is localized to the hook. These two fractures may present with pain on the ulnar half of the wrist, as well as localized swelling and tenderness. They may occur independently or simultaneously. Fractures of the hook of the hamate are strongly suspected when a deep pain is present on the ulnar half of the wrist in baseball, tennis, and squash players, and in golfers. Fractures of the hamate are most likely secondary to a direct blow by the counterforce to the butt end of a baseball bat, a golf club, or a racket. The nondominant hand is usually involved in baseball players and golfers, whereas the dominant hand is more likely involved in racquetball and tennis players. In some cases, this fracture may occur after a fall onto the base of the hypothenar area. On the PA radiograph, hamate hook fractures are suggested by absence of the hook, a cortical ring sign, or sclerosis in the region of the hook. In general, fractures of the hamate hook are often difficult to recognize by standard PA and lateral radiographs. Three specialized views—the carpal tunnel view (►**Fig. 3a**), a supinated oblique view with the wrist extended (►**Fig. 3b**), and a lateral view projected through the first web space with the thumb abducted—may bring the hamate into better visualization. The carpal tunnel view is typically used to profile the hook of the hamate, but it may be difficult in patients with acute pain who cannot fully extend their wrist. Conventional CT scan (►**Fig. 4a, b**) and three-dimensional (3D) CT (►**Fig. 4c**) are superior to conventional radiographs. CT was found to have sensitivity of 100%, specificity of 94%, and accuracy of 97.2% in defining hook fractures as compared with 72% sensitivity, 88% specificity, and 80.5% accuracy[11] in radiographs. Fracture of the body of the hamate is divided into four major groups: proximal pole, medial tuberosity, sagittal oblique, and dorsal coronal. Proximal pole fracture is due to shear forces in a fracture-dislocation of the wrist and is usually a small intra-articular osteochondral fracture. Fracture of medial tuberosity results from a direct blow to the ulnar side of the wrist. Sagittal oblique fracture is a result of high-energy impact, and dorsal coronal fracture is a result of axial impaction forces transmitted from metacarpal bones. Although most of these fractures can be identified by PA and lateral view radiographs, careful radiological examinations are needed when the fracture line lies oblique.

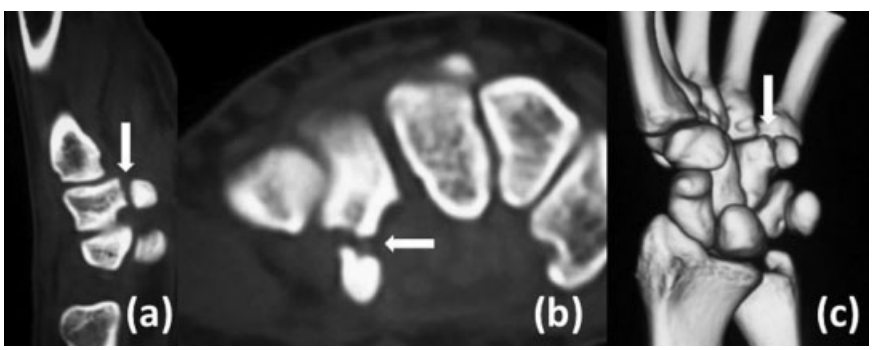


Figure 4 Computed tomography (CT) of the left hamate hook fracture in a 26-year-old male baseball player: (a, b) conventional CT and (c) three-dimensional CT. Fracture line (arrows) is more apparent compared with radiographs.

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Figure 5 Posteroanterior view of radiograph of the metacarpal base fracture of the left little finger in a 24-year-old male judo player. Intra-articular fracture (arrows) is apparent.

Metacarpal Base Fracture of the Little Finger

Metacarpal base fracture of the little finger can be classified as an intra-articular fracture of the hamate-metacarpal joint. This injury results from a longitudinally directed force along the long axis of fifth metacarpal bone, a direct blow to the hand, and rarely a fall on an outstretched hand. Although most of these fractures are easily detected by conventional radiographs (►**Fig. 5**), the extent of the injury is frequently underestimated. Bora and Didizian[12] recommended a PA view with the forearm supinated 60 degrees from the neutral position because bony overlap was eliminated and the small carpometacarpal joint surfaces clearly seen. In some cases, the diagnosis may be assisted by CT scan.

Pisiform Fracture

Pisiform fracture is a relatively rare injury among carpal fractures and accounts for ~2.0% of all carpal bone fractures.[13] Although the most common mechanism of injury is a direct blow to the hypothenar eminence, repetitive trauma has also been implicated because the pisiform is a sesamoid contained with the FCU tendon. The pisiform is so tiny that it is difficult to detect a fracture by conventional wrist radiographs. Special views such as supination oblique, carpal tunnel view, or CT (►**Fig. 6**) are useful to diagnose them.

Triangular Fibrocartilage Complex Injury

The TFCC is on the ulnar side of the wrist and plays a dual role by stabilizing DRUJ and by supporting the compression load of the carpal bones. It consists of the disc proper, meniscus homologue, ulnolunate ligament, ulnotriquetral ligament, triangular ligament (TL), and ulnar collateral ligament. In addition, TL is composed of volar and dorsal radioulnar

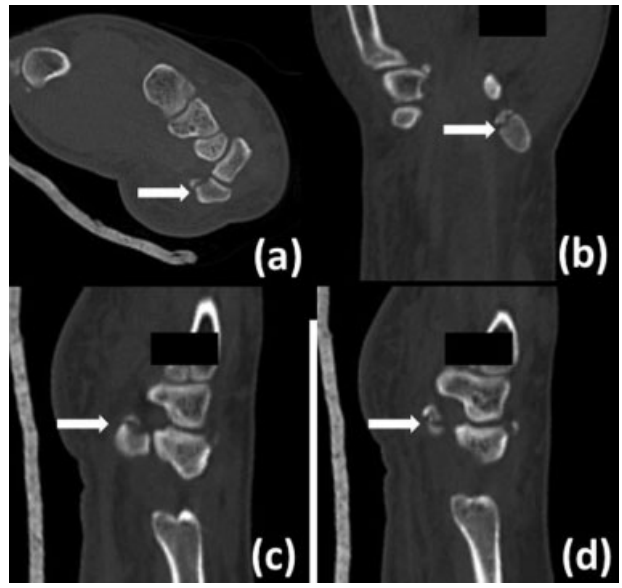


Figure 6 Computed tomography (CT) of the pisiform fracture in a 32-year-old female tennis player. Even a tiny fracture (arrows) can be detected by CT scan. (a) axial, (b) coronal, and (c, d) sagittal view.

ligaments. Traumatic injuries of the TFCC result from a forced axial load to the wrist in an extension-pronation position, such as in a fall on an outstretched hand. Another mechanism of TFCC injury may occur from a distraction force applied to the volar forearm or wrist, frequently seen in racket sports. Ulnar-positive variance is an anatomical risk factor for TFCC injury.

Radiological examinations include plain radiographs with PA view (►**Fig. 7a**) with the wrist in the neutral position and lateral view (►**Fig. 7b**). These images enable us to assess the carpal alignment, ulnar styloid morphology, adaptation of ulna to DRUJ, and, most importantly, ulnar variance. Positive ulnar variance may be linked with chondromalacia or cystic lesion in the ulnar half of the lunate. However, because TFCC is a fibrocartilage complex, the plain radiographs cannot

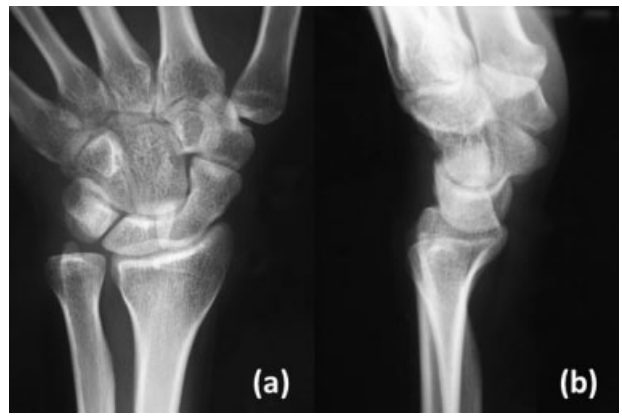


Figure 7 Radiograph of the wrist for triangular fibrocartilage complex injury in a 30-year-old male arm wrestler: (a) posteroanterior (PA) view and (b) lateral view. Although ~2 mm of ulnar-positive variance and dissociation of distal radioulnar joint (DRUJ) are apparent in the PA view, there is no subluxation of DRUJ in the lateral view. Chondromalacia of the lunate is possible.

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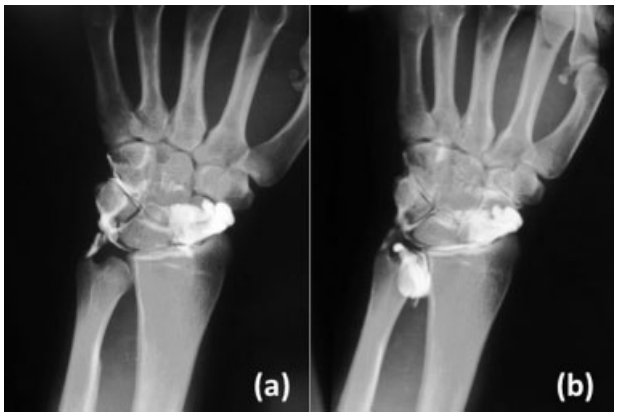


Figure 8 Arthrography of the wrist for triangular fibrocartilage complex (TFCC) injury in a 28-year-old male tennis player. Contrast medium leaks from (a) radiocarpal joint to the peripheral side of TFCC and from (b) distal radioulnar joint to the fovea of ulna, where the attachment area of the triangular ligament (TL) is located. Detachment of the TL is highly suspected.

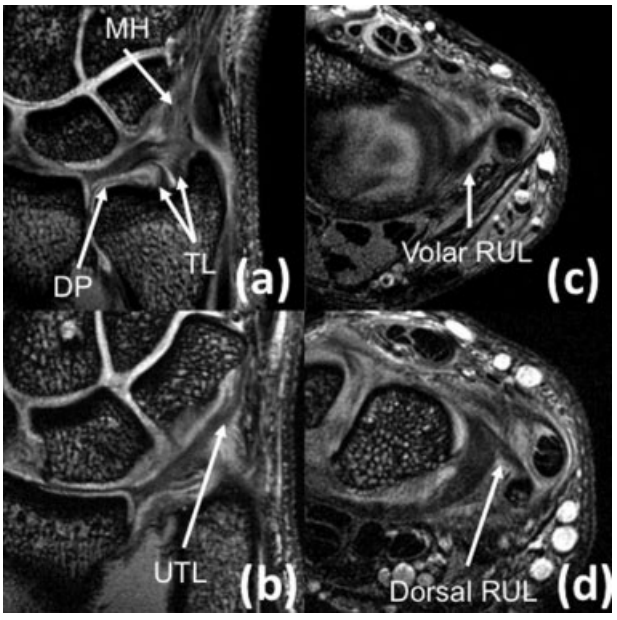


Figure 9 Normal anatomy of triangular fibrocartilage complex (TFCC) in high-resolution MRI. DP, disc proper; MH, meniscus homologue; RUL, radioulnar ligament; TL, triangular ligament; UTL, ulnotriquetral ligament. (From Tanaka T, Ogino S, Yoshioka H. Ligamentous injuries of the wrist. Semin Musculoskelet Radiol 2008; 12:359–377. Reproduced with permission of Thieme Medical Publishers, New York, NY.)

directly visualize the TFCC. To evaluate the TFCC, arthrography, which is a radiograph after injection of contrast medium to the radiocarpal joint (RCJ) or the DRUJ, is a dynamic diagnostic tool (►**Fig. 8a, b**). When the injected contrast leaks from the DRUJ to the RCJ or vice versa, detachment of the TL from the fovea of the ulna is highly suspected (►**Fig. 8b**). MRI has recently become the most useful diagnostic tool for TFCC injuries. Because these structures are located in small areas, high-resolution MRI (►**Fig. 9**) is essential to assess TFCC injuries (►**Fig. 10**).[14,15] Tanaka et al reported high-resolution MR images obtained using a microscopy surface coil allowed assessment of each TFCC component and showed high accuracy in diagnosing radial attachment injury, disc injury, and triangular ligament injury of the TFCC (100% sensitivity; 100% specificity).[14] However, conventional MR imaging sometimes does not adequately demonstrate the peripheral ulnar attachment of the TFCC because it is often obscured by an intermediate signal surrounding loose vascular connective tissue on proton-density-weighted images and T2*-weighted images.[16] MR arthrography with fat-suppressed T1-weighted images may provide better detection of peripheral TFCC tears than conventional MR imaging.

Ulnar Impaction Syndrome

Ulnar impaction syndrome is caused by the impaction between the ulnar carpal bone and the ulnar head. This can lead to ulnar-sided wrist pain as well as a degenerative/osteoarthritic condition of the ulnar side of the wrist related to excessive load-bearing across the ulnar carpus, TFCC, and ulnar head.[17] A radiograph of ulnar impaction syndrome most commonly shows positive ulnar variance. However, it may also show neutral or negative ulnar variance. Other indirect signs of ulnar impaction syndrome include subchondral changes that may include sclerosis or cystic change at the ulnar aspect of the proximal lunate, triquetrum, and ulnar head.[18] MRI is useful to detect not only the anatomical abnormality of the bone, but also secondary changes in the ulnar impaction syndrome such as degeneration or defect of cartilage, bone marrow edema, subchondral cysts, and sclerotic changes at the affected site.[16] These changes are usually observed much earlier in MR imaging than in radiography.

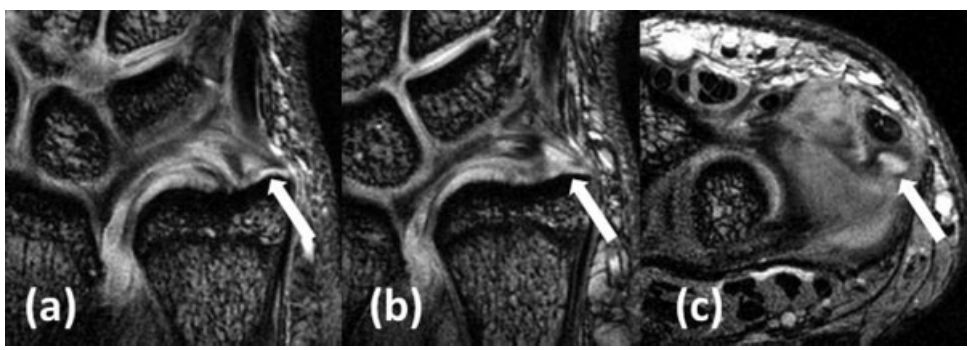


Figure 10 High-resolution MRI of the wrist for triangular fibrocartilage complex injury in an 18-year-old female judo player. There is the detachment of the triangular ligament on (a, b) coronal views and the volar radioulnar ligament on (c) axial view (arrows).

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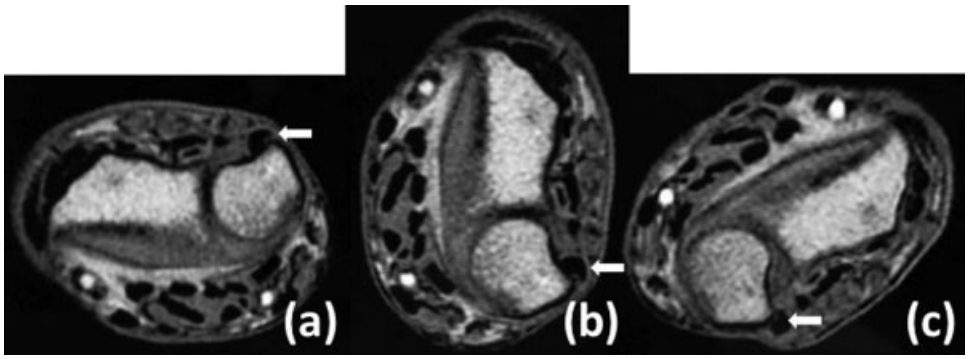


Figure 11 T1-weighted axial MRI of the wrist for extensor carpi ulnaris (ECU) dislocation in a 22-year-old male boat rower. Although the ECU is held within the groove of the ulnar head in the (a) pronation and (b) neutral position, it is dislocated ulnarly in the (c) supination position (arrows).

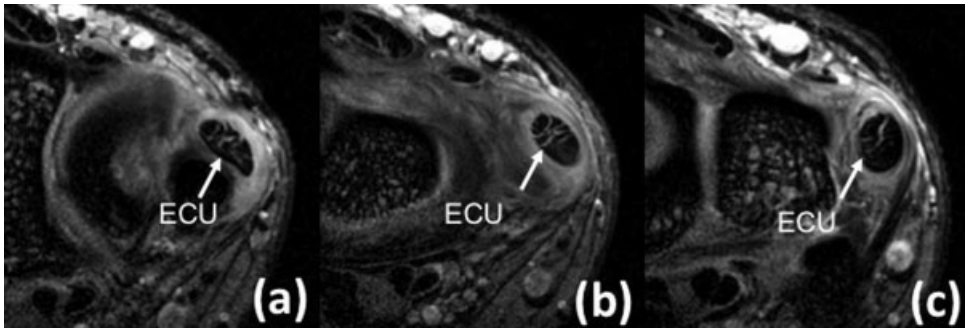


Figure 12 T2*-weighted axial MRI of the wrist for extensor carpi ulnaris (ECU) tenosynovitis in a 22-year-old female tennis player. Axial MRIs (a–c) demonstrate thickening of the ECU with associated high signal around it (arrows).

Extensor Carpi Ulnaris Disorder

The ECU passes through the sixth extensor compartment and is held within the groove of the ulnar head by a deep retinaculum, referred as the ECU subsheath or fibro-osseous band. ECU tendinitis/subluxation is seen in sports requiring repetitive wrist motion, such as rowing or racket sports that require a snap of the wrist. Repetitive stress on the ECU subsheath causes stenosing synovitis, tearing the tendon sheath, or subluxation/dislocation of ECU (►**Fig. 11**). The differential diagnosis of ECU tendinitis is a TFCC injury. Because the subsheath of the ECU is an important stabilizer of the TFCC, it is difficult to differentiate ECU disorders from traumatic tears of the TFCC by clinical assessment. We cannot detect ECU disorders by conventional radiograph because it lacks osseous abnormalities. However, MRI shows inflammation or edema around the ECU tendon with high soft tissue contrast and is a useful clinical imaging tool. Tendinopathy, tenosynovitis, and tendon rupture can be appreciated by MRI. In tendinopathy, the tendon is thickened and its signal is increased in both T1- and T2-weighted images as well as T2*-weighted images (►**Fig. 12**). Fluid around the tendon within the tendon sheath suggests an active tenosynovitis.

Flexor Carpi Ulnaris Disorder

One of the most popular FCU disorders in athlete is FCU tendinitis, considered to result from repetitive ulnar deviation motion such as hammering. FCU tendinitis has been diagnosed in squash, badminton, and golf players. Repetitive high-speed impacts with the ball are the mechanism of injury.[19] FCU tendinitis is seen in a variety of sports in which athletes use their hands frequently. However, it may also appear without any apparent impact. We can sometimes detect FCU tendinitis by tangential radiographs of the volar, ulnar aspect of the wrist that show calcium deposition. MR imaging reveals signs of tendinitis with increased signal on T1- and T2-weighted images and can also demonstrate calcification deposition in calcific tendinitis, especially with gradient-echo sequences.

Summary

Precise history regarding the mechanism of injury is essential to diagnose accurately an athlete's ulnar-sided wrist pain. Osseous injuries, ligamentous and tendinous disorders, and ulnar impaction with or without TFCC injury are the main differential diagnosis. Comprehensive radiological evaluations including radiographs, CT, and MRI are needed.

References

1 Dawson WJ, Pullos N. Baseball injuries to the hand. Ann Emerg Med 1981;10(6):302–306
2 Winge S, Jørgensen U, Lassen Nielsen A. Epidemiology of injuries in Danish championship tennis. Int J Sports Med 1989;10(5): 368–371
3 Hutchinson MR, Laprade RF, Burnett QM II, Moss R, Terpstra J. Injury surveillance at the USTA Boys' Tennis Championships: a 6-yr study. Med Sci Sports Exerc 1995;27(6):826–830

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4 McCarroll JR. The frequency of golf injury. Clin Sports Med 1996; 15(1):1–7
5 Gabel GT. Gymnastic wrist injuries. Clin Sports Med 1998; 17(3):611–621
6 Caine D, Roy S, Singer KM, Broekhoff J. Stress changes of the distal radial growth plate. A radiographic survey and review of the literature. Am J Sports Med 1992;20(3):290–298
7 DiFiori JP, Puffer JC, Mandelbaum BR, Mar S. Factors associated with wrist pain in the young gymnast. Am J Sports Med 1996; 24(1):9–14
8 Mandelbaum BR, Bartolozzi AR, Davis CA, Teurlings L, Bragonier B. Wrist pain syndrome in the gymnast. Pathogenetic, diagnostic, and therapeutic considerations. Am J Sports Med 1989;17(3):305–317
9 Fyrkman G. Fracture of the distal radius including sequelae—shoulder-hand-finger syndrome, disturbance in the distal radio-ulnar joint and impairment of nerve function. A clinical and experimental study. Acta Orthop Scand Suppl 1967;(Suppl 108):3
10 Nakamura T, Peimer CA, Moy OJ, et al. Relationship between fracture of the ulnar styloid and DRUJ instability: a biomechanical study. J Bone Joint Surg Am. In press
11 Andresen R, Radmer S, Sparmann M, Bogusch G, Banzer D. Imaging of hamate bone fractures in conventional X-rays and high-resolution computed tomography. An in vitro study. Invest Radiol 1999;34(1):46–50
12 Bora FW Jr, Didizian NH. The treatment of injuries to the carpometacarpal joint of the little finger. J Bone Joint Surg Am 1974; 56(7):1459–1463
13 McCarty V, Farber H. Isolated fracture of the pisiform bone. J Bone Joint Surg Am 1946;28:390
14 Tanaka T, Yoshioka H, Ueno T, Shindo M, Ochiai N. Comparison between high-resolution MRI with a microscopy coil and arthroscopy in triangular fibrocartilage complex injury. J Hand Surg Am 2006;31(8):1308–1314
15 Yoshioka H, Tanaka T, Ueno T, et al. Study of ulnar variance with high-resolution MRI: correlation with triangular fibrocartilage complex and cartilage of ulnar side of wrist. J Magn Reson Imaging 2007;26(3):714–719
16 Watanabe A, Souza F, Vezeridis PS, Blazar P, Yoshioka H. Ulnar-sided wrist pain. II. Clinical imaging and treatment. Skeletal Radiol 2010;39(9):837–857
17 Cerezal L, del Piñal F, Abascal F, García-Valtuille R, Pereda T, Canga A. Imaging findings in ulnar-sided wrist impaction syndromes. Radiographics 2002;22(1):105–121
18 Imaeda T, Nakamura R, Shionoya K, Makino N. Ulnar impaction syndrome: MR imaging findings. Radiology 1996;201(2): 495–500
19 Werner SL, Plancher KD. Biomechanics of wrist injuries in sports. Clin Sports Med 1998;17(3):407–420

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